

LEVEL 3 PROJECTS (Part 2)

Instrument Connectivity Language (ICL)

- Language used to describe test instruments and their access network.
- ICL is not a complete netlist. It only consists of abstract information required to navigate the instrument access network.

Procedural Description Language (PDL)

- PDL provides a way to define procedures to operate an instrument.
- PDL is a sequence of commands that are a set of stimuli and expected responses that are applied on the interface of an instrument.
 - **iWrite** : Stimuli to the instrument are specified by iwrite.
 - Ex: iWrite Block1.MyTdr.R 8b00000101
 - In this example, the register R, an object in ICL instance MyTdr that is instantiated in instance Block1, is loaded with a value of 8b00000101
 - **iRead** : Expected responses are specified with the command iRead.
 - Ex: iRead Block1.MyTdr.R 0b0101
 - In this example, if R is an 8 bit wide register, then the compare value is zero padded to 0b00000101

challenges in MBIST grouping :-

↳ more memories run in 11th ← more power consumption

Eg: - Controller C1 ← 50 memories

↳ 50 memories are running 11th.

Power analysis team suggestion :-

Only 25 mems should run 11th to avoid power issues in silicon.

So we can divide the memories in 2 steps.

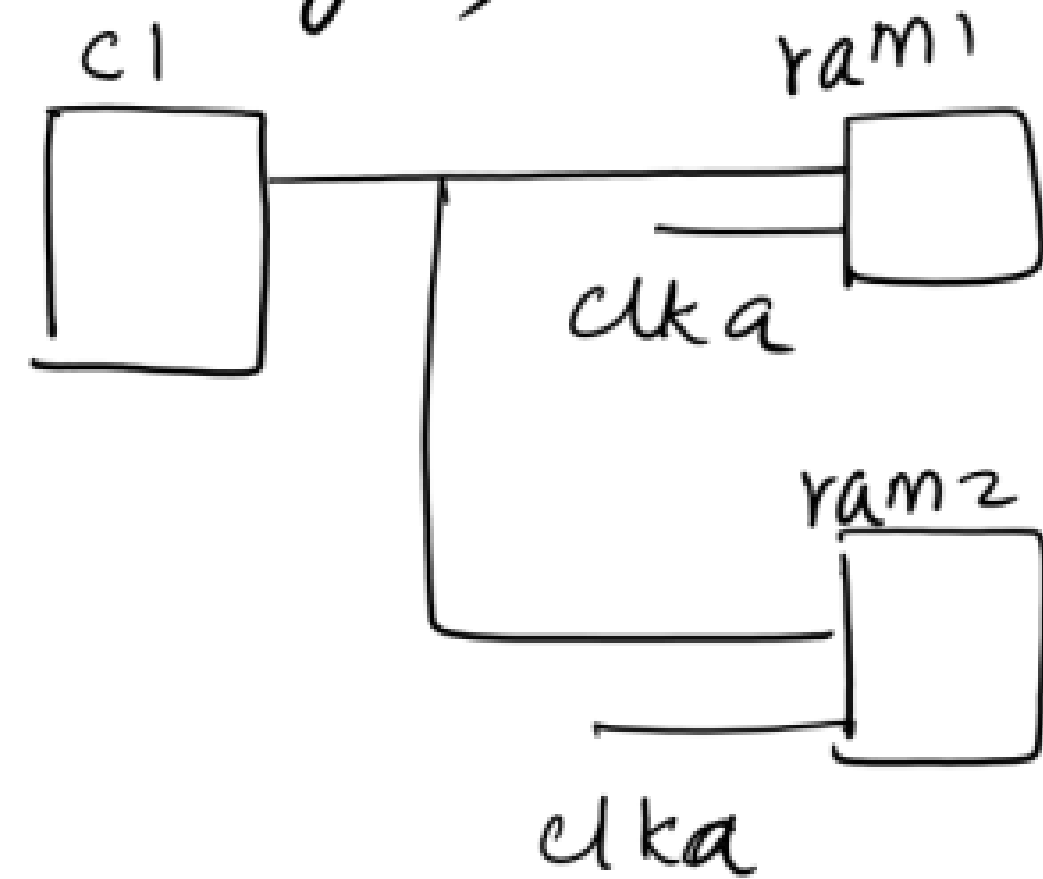
Controller C1 Step0 ← 25 memories (run parallelly)

Step1 ← 25 memories (run parallelly)

After completion of step 0, step 1 will start to run,

Procedure for dividing mems in different steps:-

Note :-
By editing our original design,



This is just done to show how to divide mems in diff steps.

PROCEDURE:-

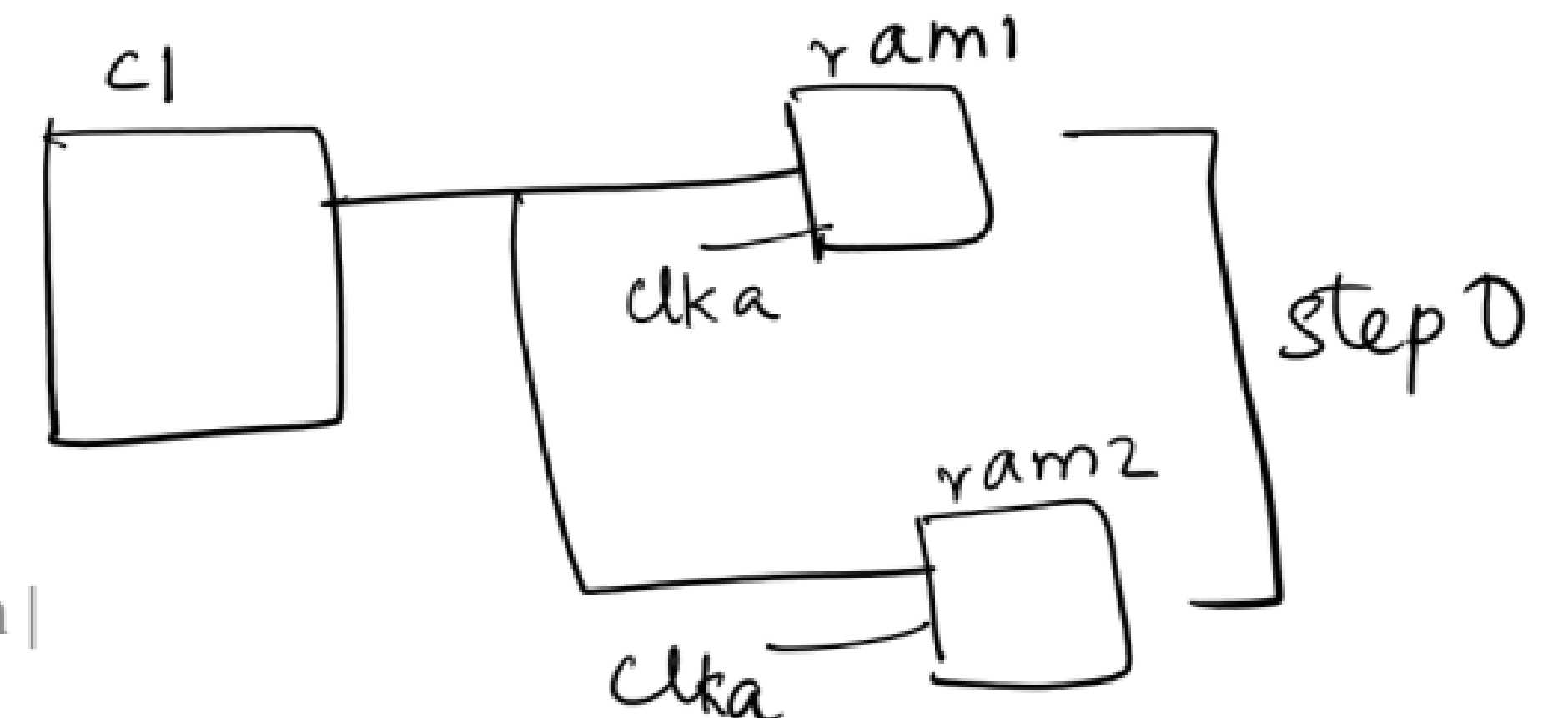
1. after create dft-spec command, write out the DFT spec

Command :- write-config-data <name of the file>

```
MemoryBist {
  ijtag_host_interface : Sib(mbist
  Controller(c1) {
    clock_domain_label : clka;
    Step {
      MemoryInterface(m1) {
        instance_name : ram1;
      }
      MemoryInterface(m2) {
        instance_name : ram2;
      }
    }
  }
}
```

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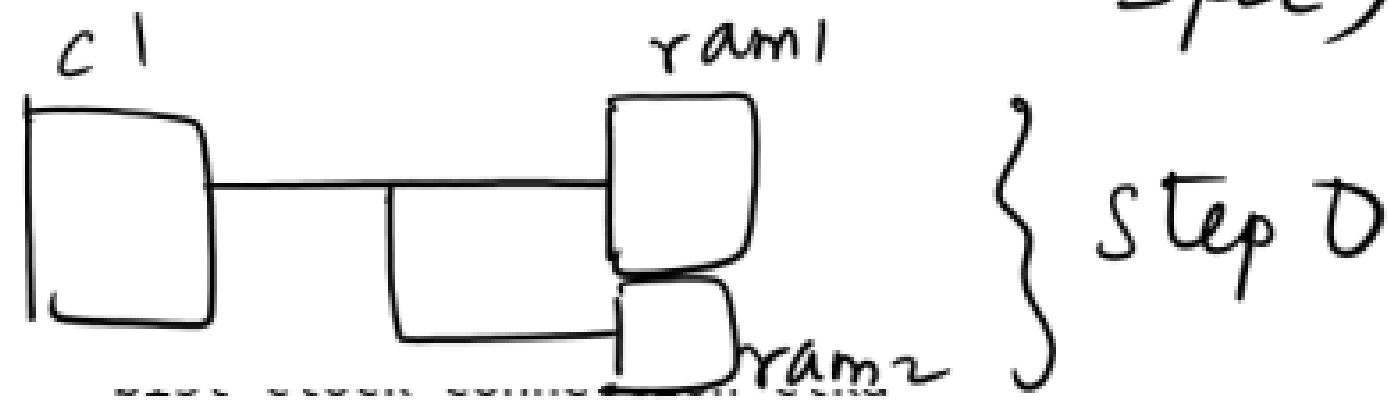


2. Edit the dft spec file

```
MemoryBist {  
  ijtag_host_interface : Sib(mbist);  
  Controller(c1) {  
    clock_domain_label : clka;  
    Step {  
      MemoryInterface(m1) {  
        instance_name : ram1;  
      }  
    }  
    Step {  
      MemoryInterface(m2) {  
        instance_name : ram2;  
      }  
    }  
  }  
}
```

3. Read the edited dft spec file
read-config-data dft-spec-jun4-edited

By default :- (without editing any spec)



```

module_name corea_first_insertion_tessent_mbist_c1_controller
design_instance corea_first_insertion_tessent_mbist_c1_controller_inst
bap_module_name corea_first_insertion_tessent_mbist_bap
bap_design_instance corea_first_insertion_tessent_mbist_bap_inst
bist_port {corea_first_insertion_tessent_mbist_bap_inst/bistEn[0]}
Step {
  0 {
    StepInfo {
      power_mw 1.66
      run_time_ns 15840.0
      test_cycles 1584
      comparator_location per_interface
    }
    MemoryInstance {
      ram1 {
        power_mw 0.83
        interface_id m1
        template /Core(SYNC_1R1W_16x8)/Memory
        type sram
        object_id instance:0
        MemoryInterfaceInfo {
          design_instance ram1_interface_inst
          module_name corea_first_insertion_tessent_mbist_c1_interface_m1
        }
        logical_ports 1R1W
        test_ports 1
      }
      ram2 {
        power_mw 0.83
        interface_id m2
        template /Core(SYNC_1R1W_16x8)/Memory
        type sram
      }
    }
  }
}
  
```

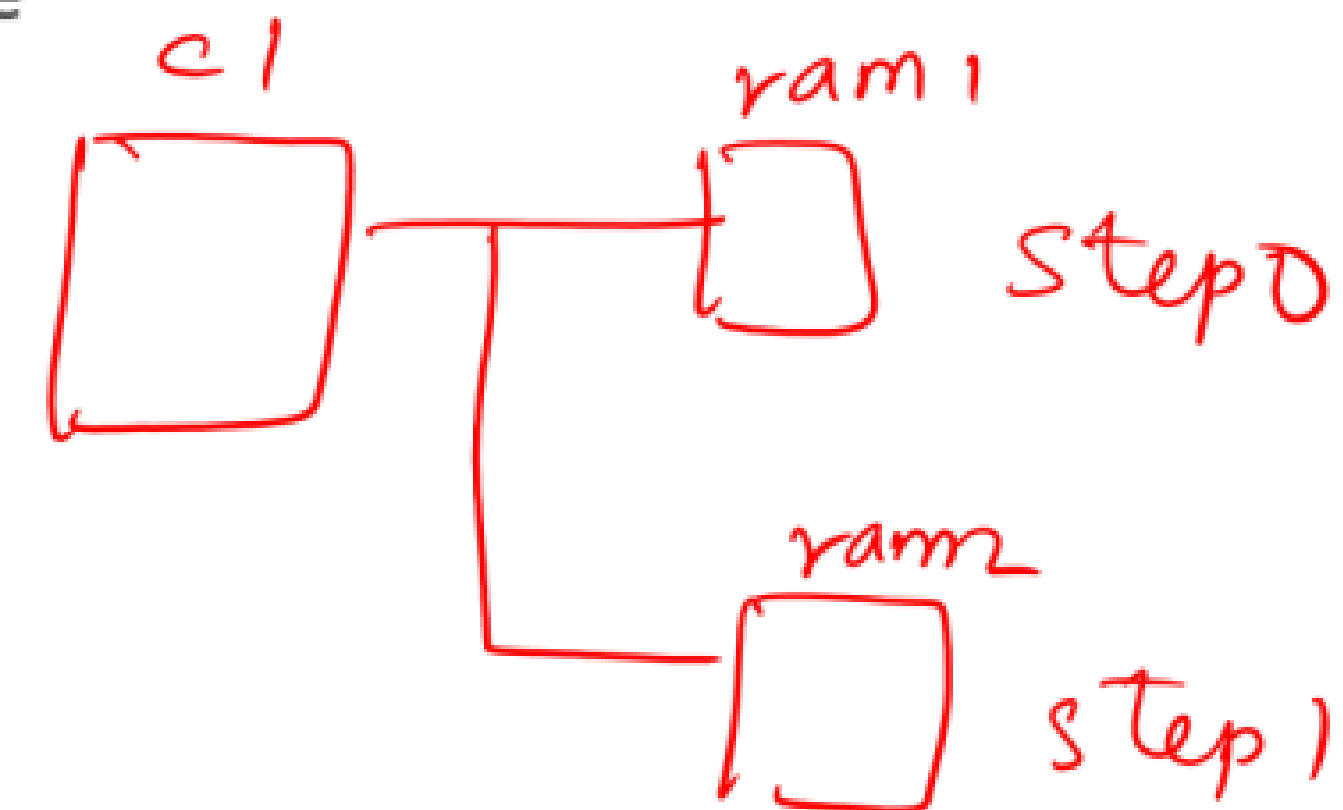
After editing the default spec

```

module_name corea_first_insertion_tessent_mbist_c1_controller
design_instance corea_first_insertion_tessent_mbist_c1_controller_inst
bap_module_name corea_first_insertion_tessent_mbist_bap
bap_design_instance corea_first_insertion_tessent_mbist_bap_inst
bist_port {corea_first_insertion_tessent_mbist_bap_inst/bistEn[0]}
  
```

```

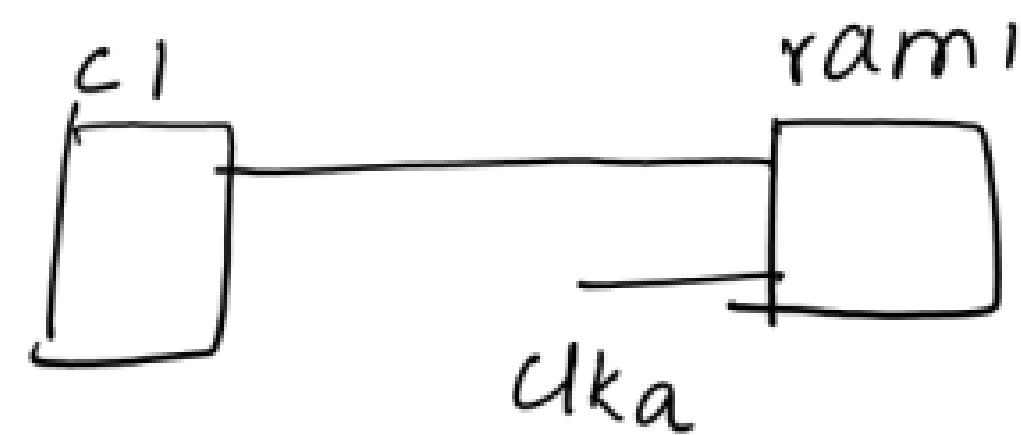
Step {
  0 {
    StepInfo {
      power_mw 0.83
      run_time_ns 15840.0
      test_cycles 1584
      comparator_location per_interface
    }
    MemoryInstance {
      ram1 {
        power_mw 0.83
        interface_id m1
        template /Core(SYNC_1R1W_16x8)/Memory
        type sram
        object_id instance:0
        MemoryInterfaceInfo {
          design_instance ram1_interface_inst
          module_name corea_first_insertion_tessent_mbist_c1_interface_m1
        }
        logical_ports 1R1W
        test_ports 1
      }
    }
  }
}
  
```



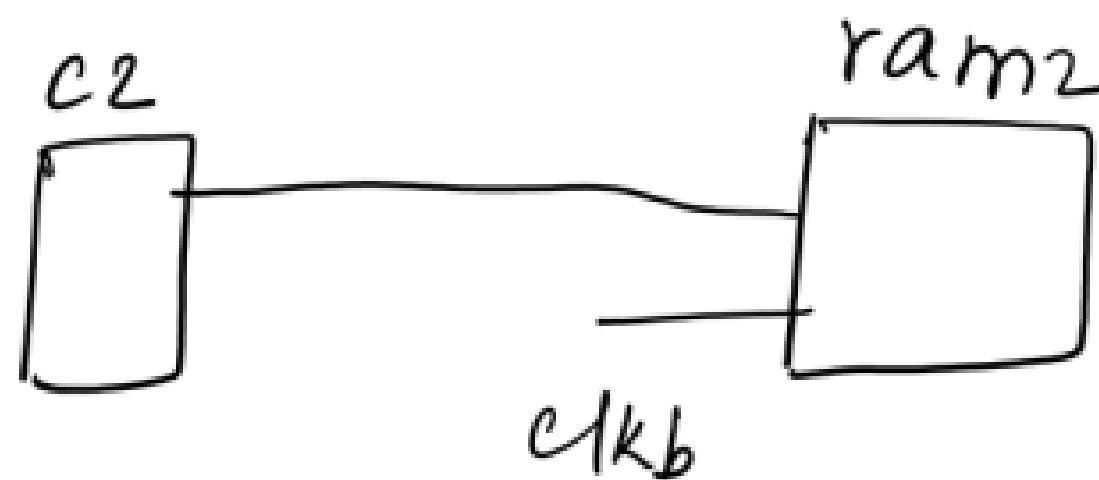
```

1 {
  StepInfo {
    power_mw 0.83
    run_time_ns 15840.0
    test_cycles 1584
    comparator_location per_interface
  }
  MemoryInstance {
    ram2 {
      power_mw 0.83
      interface_id m2
      template /Core(SYNC_1R1W_16x8)/Memory
      type sram
      object_id instance:1
      MemoryInterfaceInfo {
        design_instance ram2_interface_inst
        module_name corea_first_insertion_tessent_mbist_c1_interface_m2
      }
      logical_ports 1R1W
      test_ports 1
    }
  }
}
  
```

Our testcase :-



ram1 is controlled in step 0 of c1



ram2 is controlled in step 0 of c2

```

corea_first_insertion_tessent_mbist_c1_controller_inst {
  bist_clock_period_ns 10.0
  bist_clock_connection clka
  module_name corea_first_insertion_tessent_mbist_c1_controller_inst
  design_instance corea_first_insertion_tessent_mbist_c1_controller_inst
  bap_module_name corea_first_insertion_tessent_mbist_bap
  bap_design_instance corea_first_insertion_tessent_mbist_bap
  bist_port {corea_first_insertion_tessent_mbist_bap_inst/bist
Step {
  0 {
    StepInfo {
      power_mw 0.83
      run_time_ns 15840.0
      test_cycles 1584
      comparator_location per_interface
    }
    MemoryInstance {
      ram1 {

```

```

corea_first_insertion_tessent_mbist_c2_controller_inst {
  bist_clock_period_ns 13.2
  bist_clock_connection clkb
  module_name corea_first_insertion_tessent_mbist_c2_controller_inst
  design_instance corea_first_insertion_tessent_mbist_c2_controller_inst
  bap_module_name corea_first_insertion_tessent_mbist_bap
  bap_design_instance corea_first_insertion_tessent_mbist_bap
  bist_port {corea_first_insertion_tessent_mbist_bap_inst/bist
Step {
  0 {
    StepInfo {
      power_mw 0.63
      run_time_ns 20908.8
      test_cycles 1584
      comparator_location per_interface
    }
    MemoryInstance {
      ram2 {

```